ENGR - Lab 5 - Part C - Car Response Analysis

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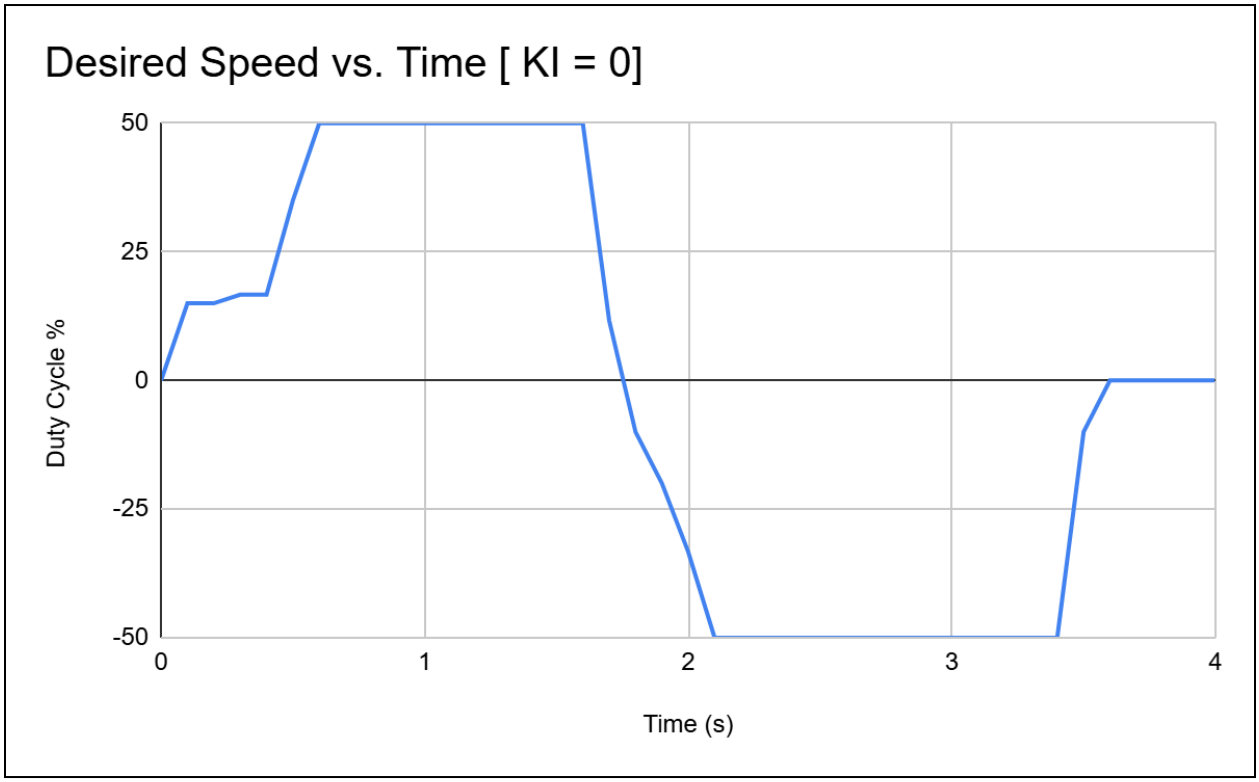
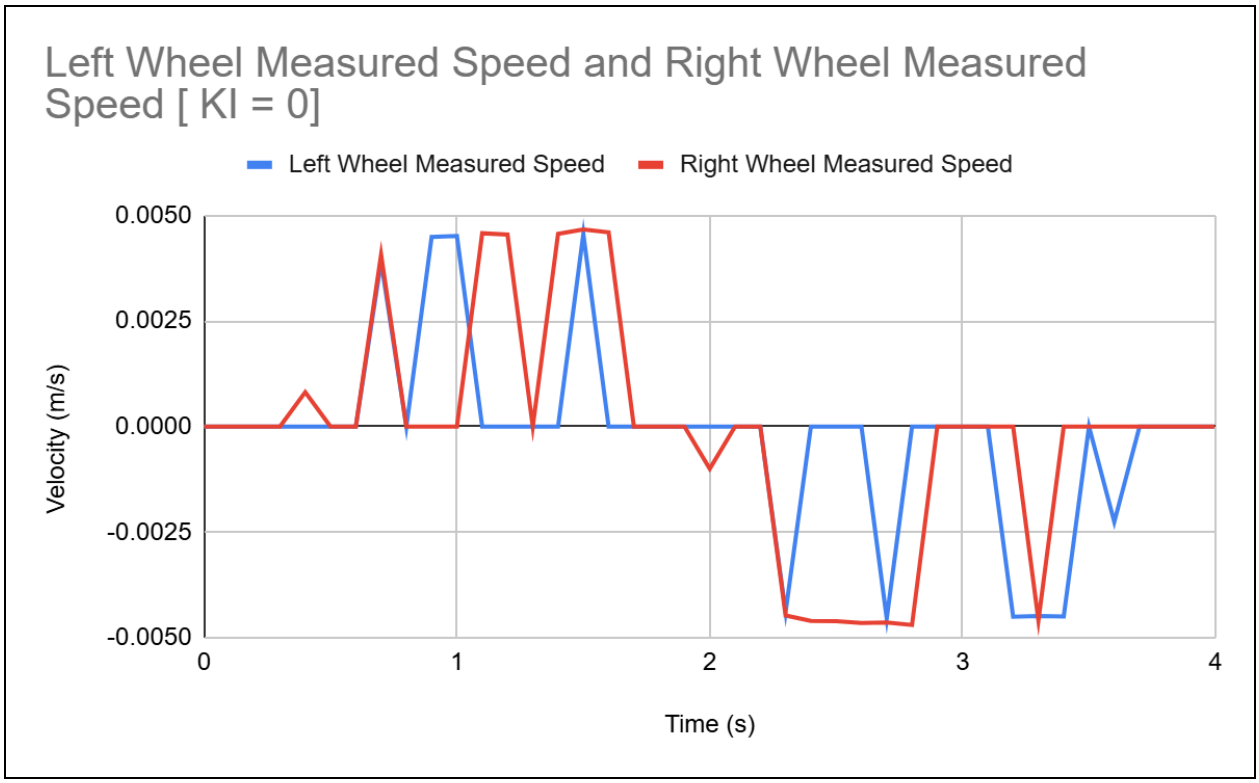
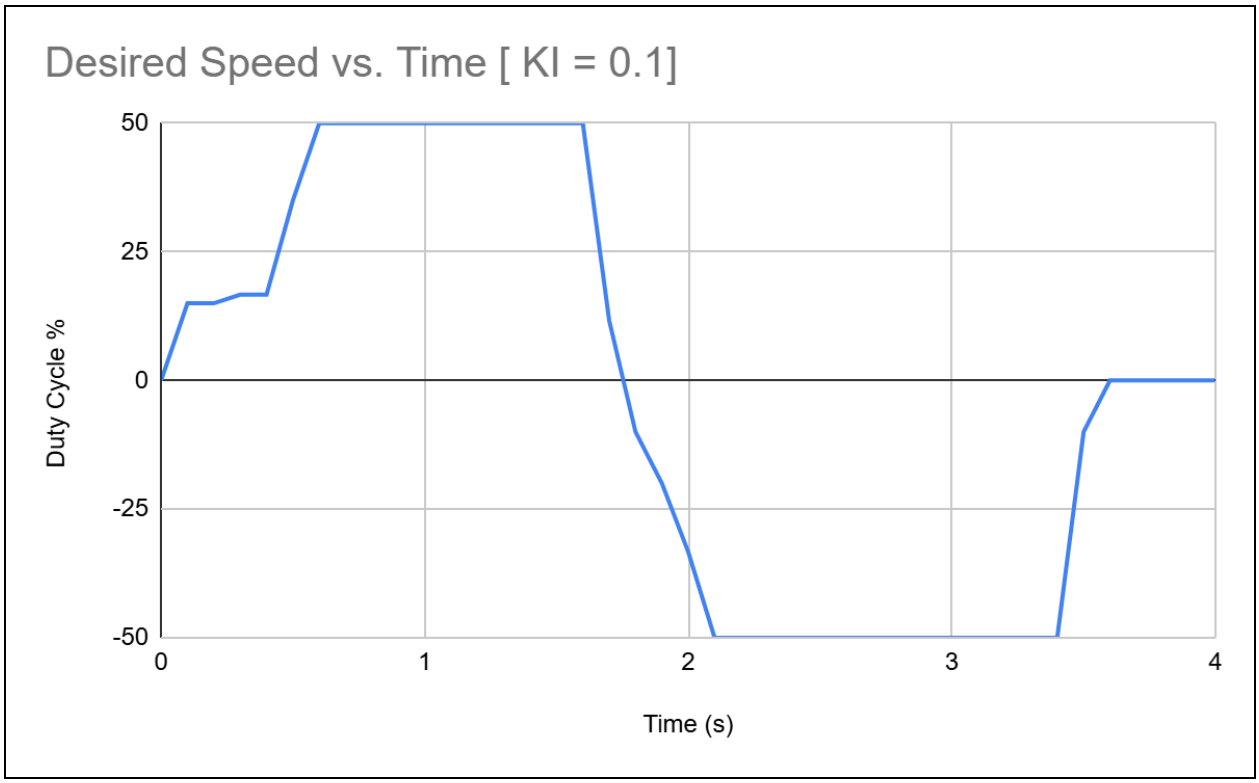
1a. Measuring Forward, Backward Speed (Ki = 0)

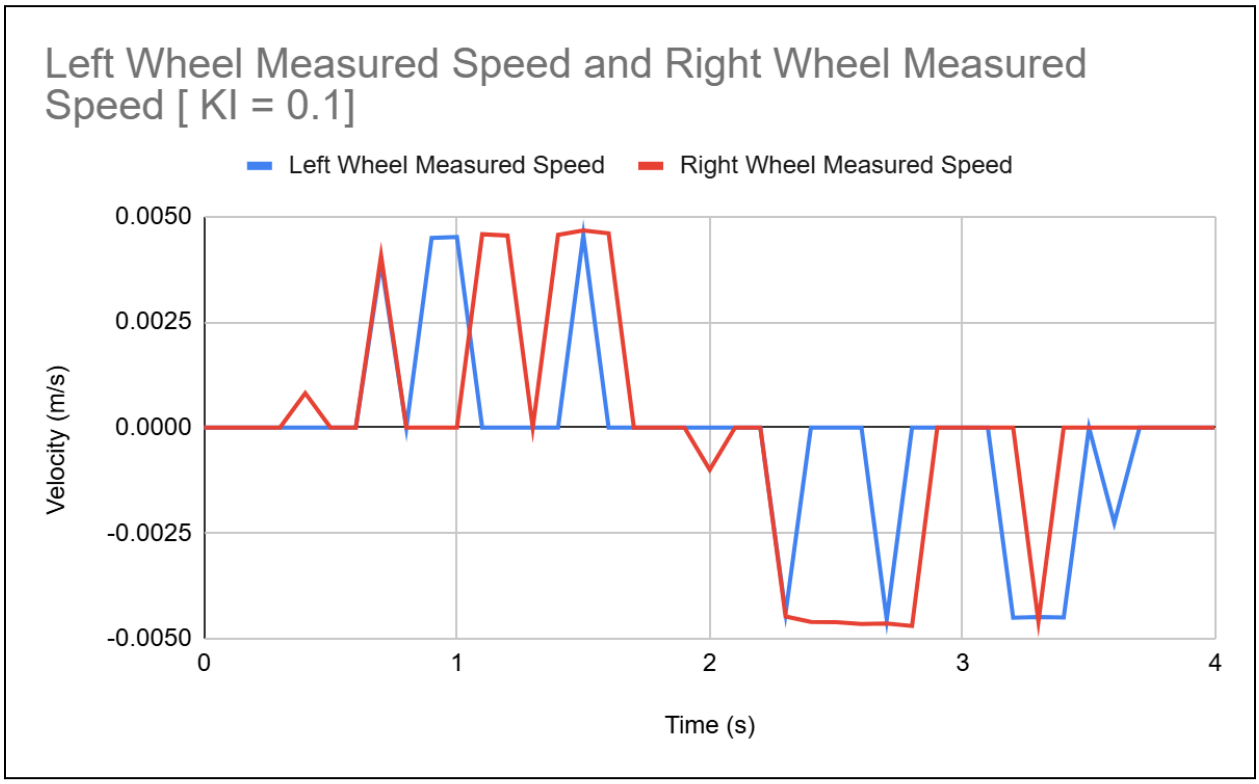




The first graph depicting Desired Speed in Duty Cycle % over time showcases the car's ability to reach max Speed, fully stop, then reach min Speed. It is not perfectly linear due to friction, handling error, and the lack of an Integral Constant. The second graph depicts the left and right wheel speeds with an Integral Constant of zero over time as the car moved forwards and backwards. Overall the shape of the desired speed graph matches that of the measured left and right speeds.

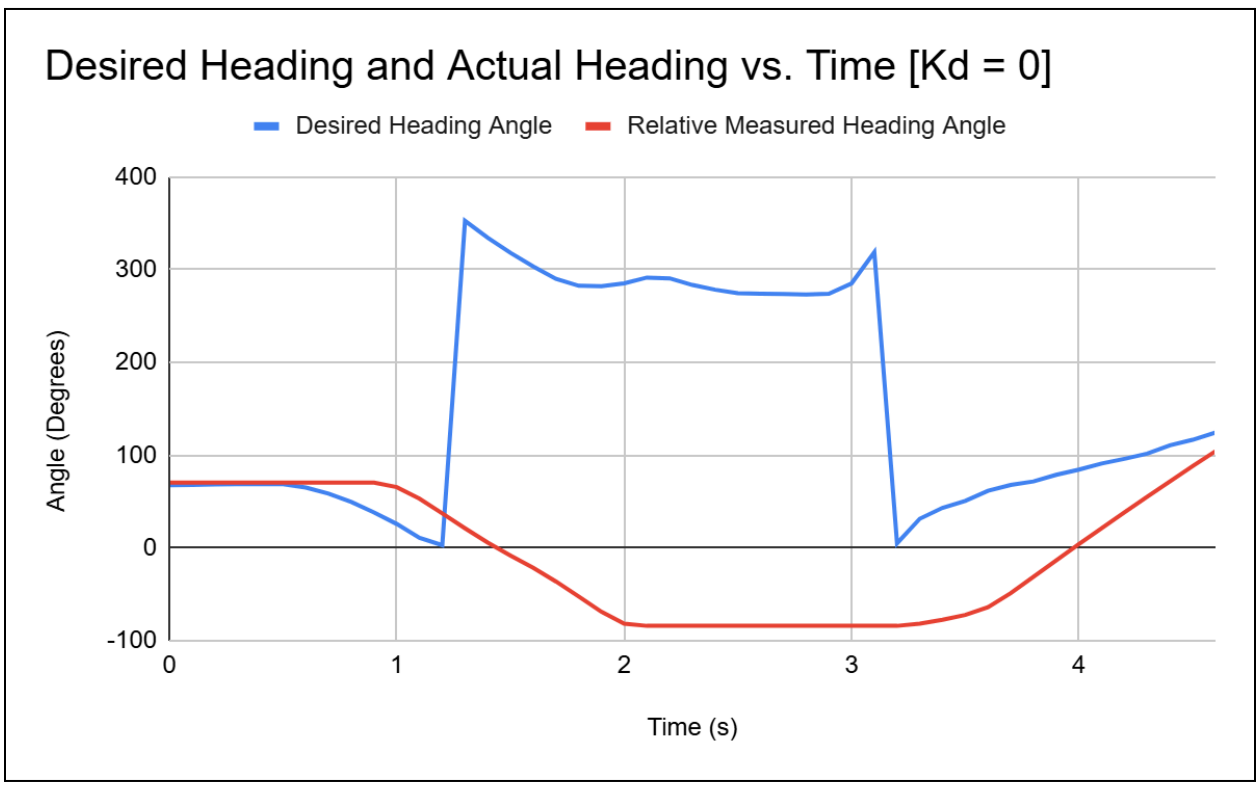
1b. Measuring Forward, Backward Speed (Ki = 0.1)

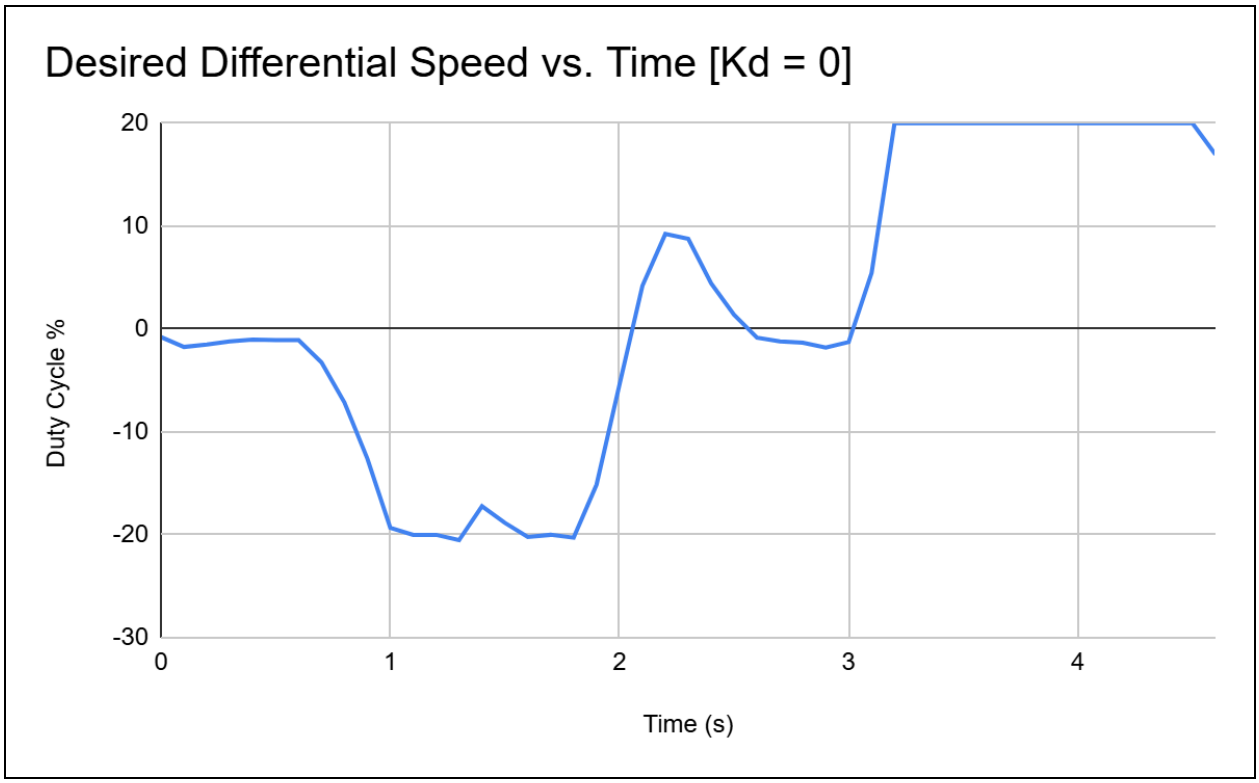

Desired Speed vs. Time [KI = 0.1]
50
25
0
-25
-50
Duty Cycle %
0
1
2
3
4
Time (s)

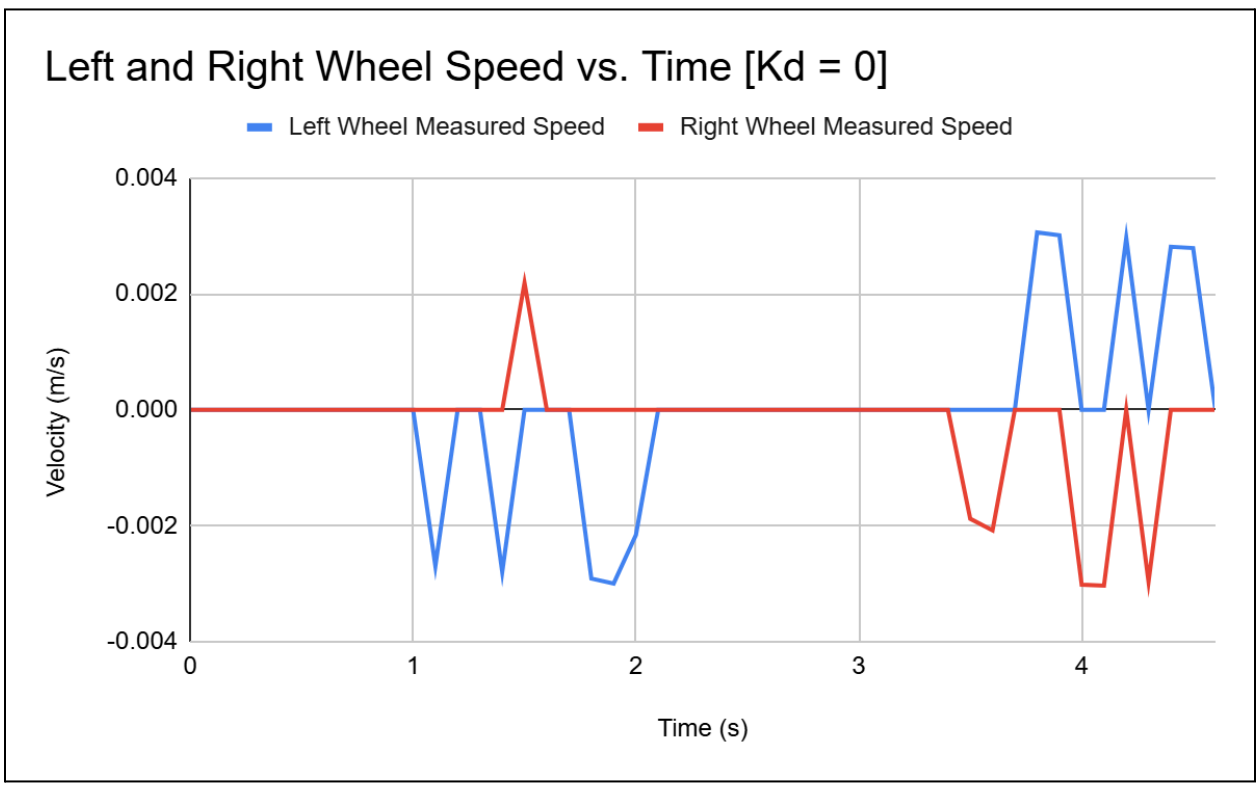

Left Wheel Measured Speed and Right Wheel Measured Speed [KI = 0.1]
Left Wheel Measured Speed
Right Wheel Measured Speed
0.0050
0.0025
0.0000
-0.0025
-0.0050
Velocity (m/s)
0
1
2
3
4
Time (s)

The first graph depicting Desired Speed in Duty Cycle % over time showcases the car's ability to reach max Speed, fully stop, then reach min Speed. This graph is slightly smoother than with Ki = 0, although a larger Ki value in part B of the lab may have been beneficial. The second graph depicts the left and right wheel speeds with an Integral Constant of zero over time as the car moves forwards and backwards. The effect of the Integral constant on the system was very minimal.

2a. Measuring Turning Speed Via Compass (Kd = 0)



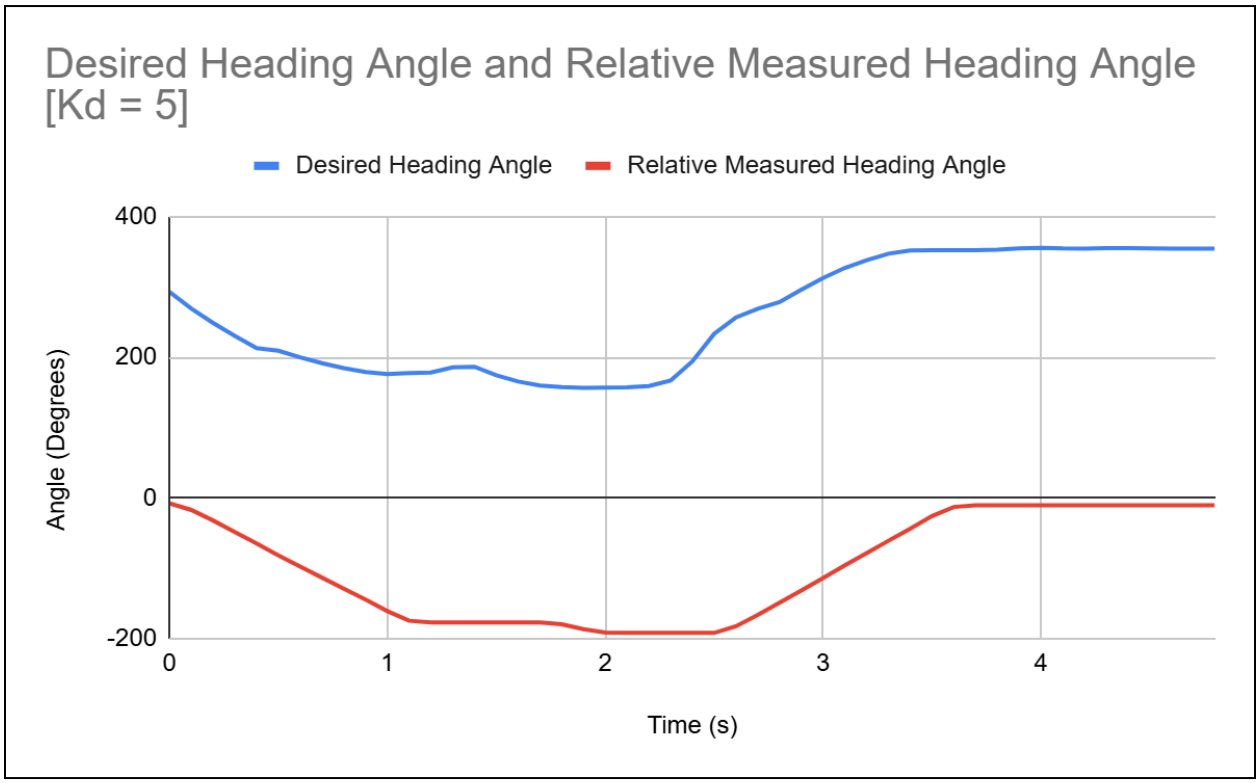


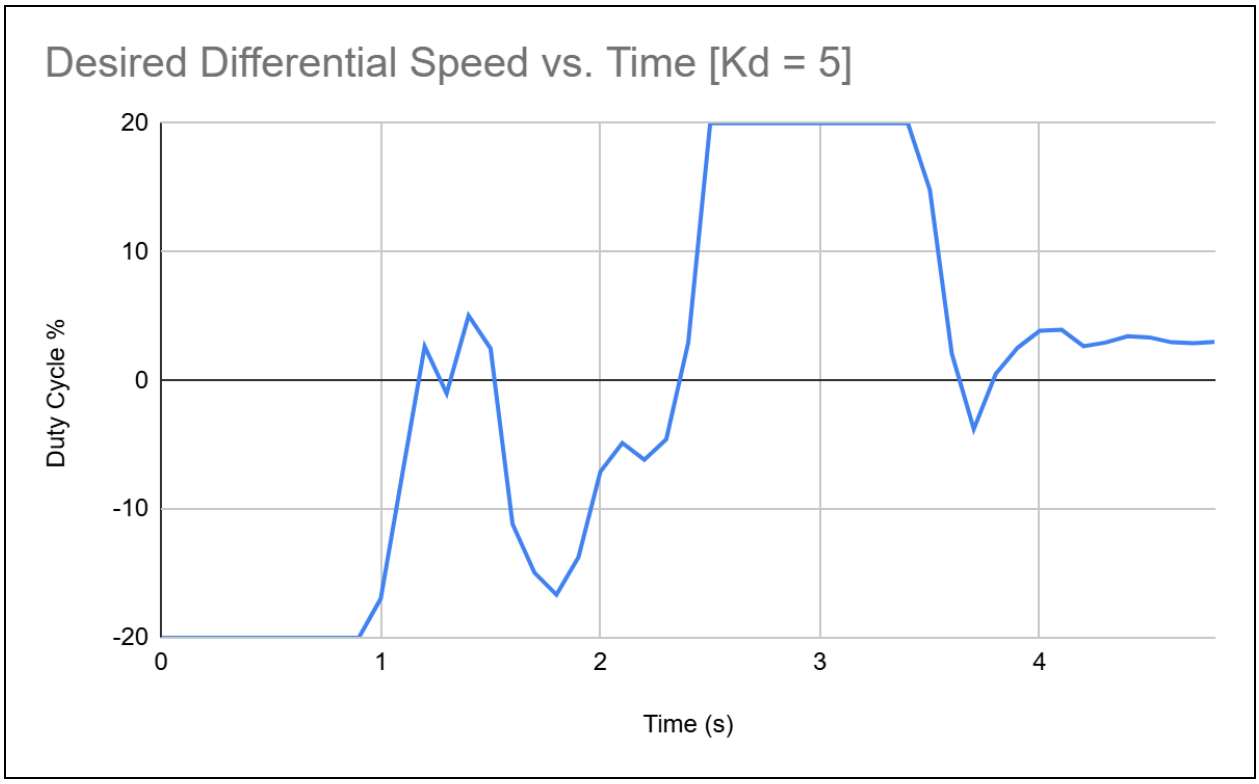


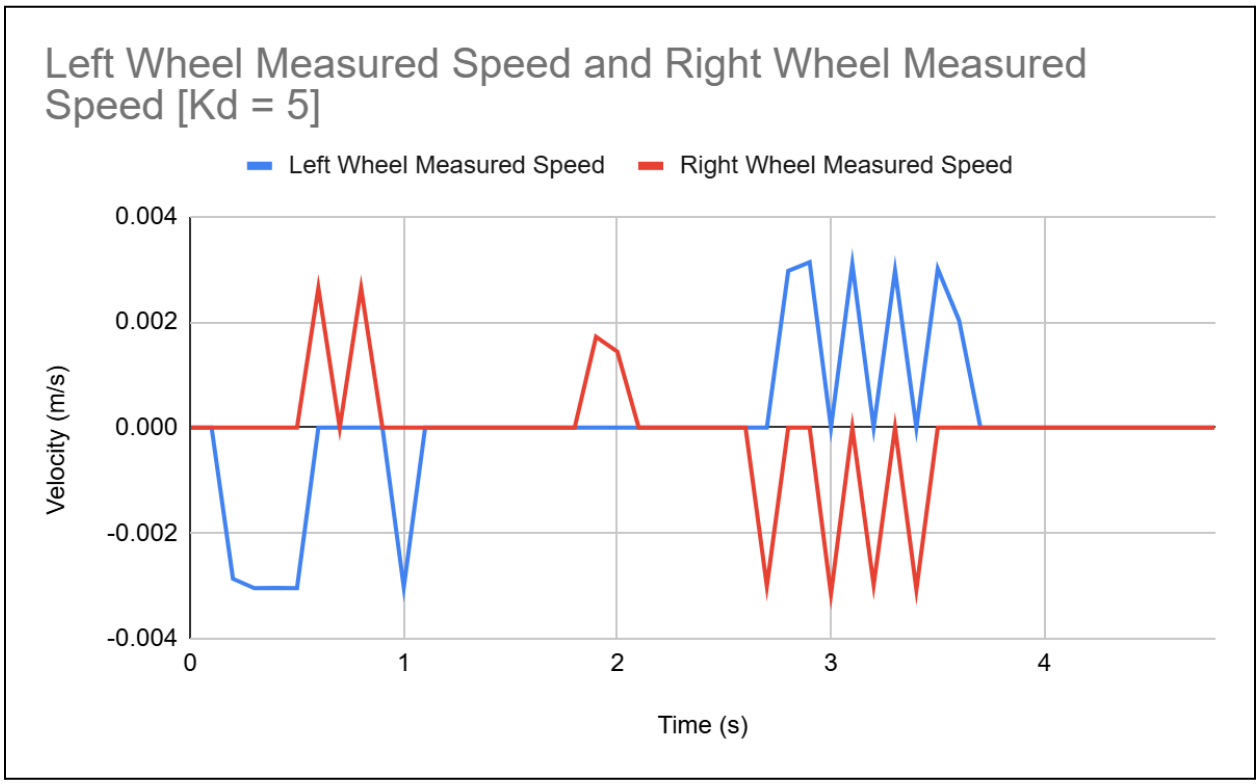
Considering the compass wrapping included in the program but not these measurements, the topmost graph appears to indicate that the car corrected quickly

to a new steady state angle measurement, although the delay to begin correction could imply that certain constants were low to begin with. This slow start may imply that the program is overdamped. Further, after one second, the limited duty cycle of differential speed kicks in, creating a steady change in angle between one and two seconds. The data appears to indicate that despite lack of finesse in outputting differential speed or measuring velocity, the car did correct to the desired heading, and based on the trends of the data at 4.5 seconds, attempted to correct remaining steady state error also.

2b. Measuring Turning Speed Via Compass (Kd = 5)



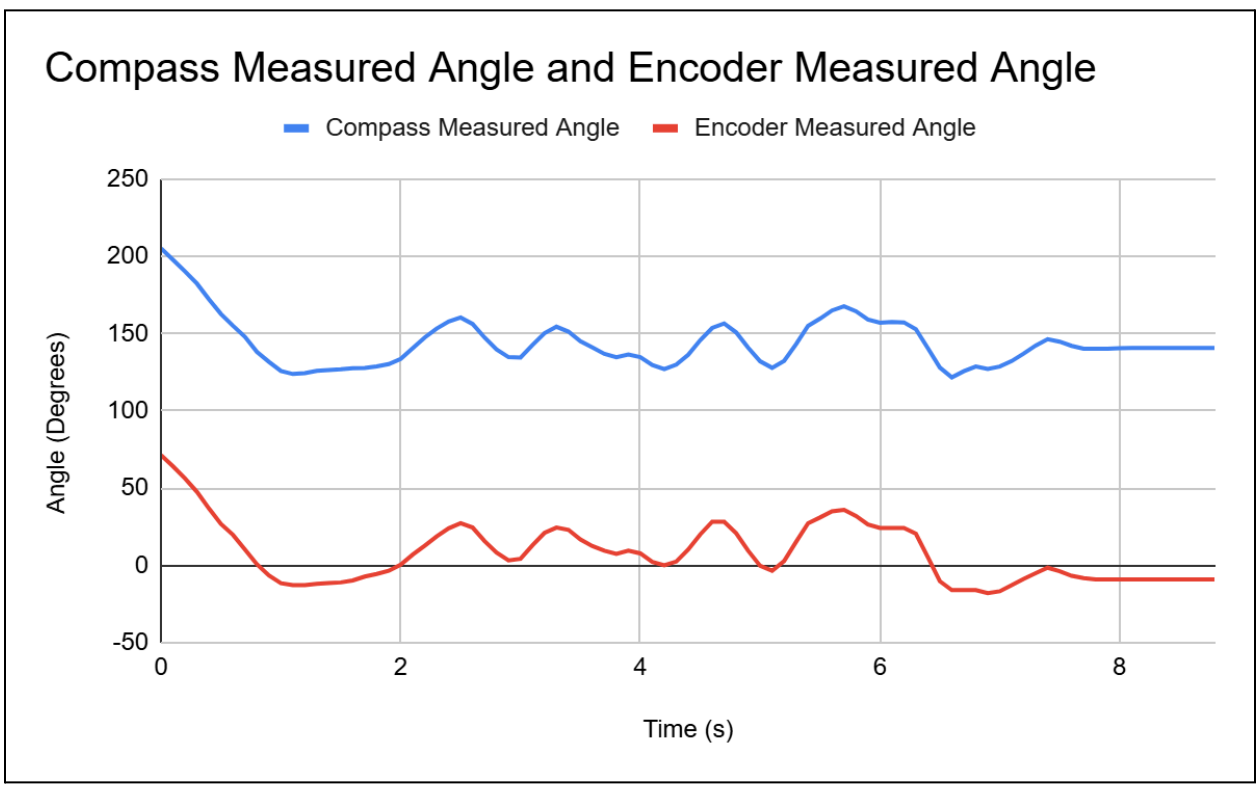




As shown in the first graph, with the addition of a derivative gain constant, the desired heading angle and relative measured heading angle are directly related.

Compared to the graph with Kd = 0, the desired matches the measured. The areas of saturation appear to indicate that the duty cycle measurement was often at maximum speed. That stated, the correspondence of small changes in desired heading corresponding to proportionally large fluctuations in the duty cycle between one and two seconds indicate that the car's ability to correct smaller errors suffers. This is supported by the fact that the velocity of the car in this section is at an absolute minimum. This could be an effect of the damping, as a higher derivative constant resists small changes like those demonstrated above. That said, if the derivative constant resists human imprecision in input (shaky hands, etc.) then the slower response time to smaller inputs is an overall boon to robot precision.

3. Compass and Encoder Heading Comparison



For the unpowered test of the compass and motor direction, there was a persistent steady state error difference between the two values. However, it is important to note that this difference is constant, and due to a slight offset of the compass with respect to the car when attached. Considering this offset, the car's encoder measurements undergo nearly proportional changes when compared to the compass. Admittedly, the difference in measurements does not remain constant forever, and towards the end of the program starts to increase. Therefore, the designed system of encoder measurement does match with the compass measurements generally, although the error between these measurements may increase slightly with time. Additional factors could be that the compass was not affixed securely enough to the car during testing, and slipped slightly, contributing to the measured offset difference.

Time (s)	Compass	Encoder	Offset
1	205.1	71.25	133.85
8.9	140.7	-9	149.7